

Glutathione peroxidase activity assessment

Material:

1. Sample
2. Potassium phosphate buffer
3. Phosphate buffer
4. EDTA
5. NADPH
6. NaN_3 100 mM
7. Glutathione reductase
8. Reduced glutathione
9. Milli-Q water
10. H_2O_2 4 mM

Procedures:

1. Homogenize the sample in a sonicator (Branson Sonifier Model 250; three 4-s cycles) in potassium phosphate buffer 100 mM and EDTA 50 mM, pH 7.4, and centrifuge the homogenate at 10.000 rpm and 4 °C for 10 min.
2. Prepare a reaction solution in 20 mL of phosphate buffer (100 mM; pH 7.0) containing EDTA 5 mM, 3.12 g of NADPH, 10 μL of glutathione reductase (500 U/mg of protein/mL), 250 μL of NaN_3 100 mM, and 7.68 g of reduced glutathione.
3. Then, add 510 μL of the reaction solution, 30 μL of milli-Q water, and 30 μL of homogenate to the optical cell.
4. The enzymatic reaction is initiated with the addition of 60 μL of H_2O_2 4 mM.
5. Measure NADPH conversion to NADP in a spectrophotometer (Hitachi, model U-2001, Hitachinaka City, Japan) for 5 min.
6. The enzyme unit is determined as the oxidation of 1 mol of NADPH per minute and is calculated on the basis of the NADPH absorbance at 340 nm.
7. Normalize the results by the protein content in the sample and expressed as a percentage of the placebo group.

